Recitation 3: MIT 6.041 Probabilistic Systems Analysis

1. $H_1 = \{1st\ toss\ is\ a\ head\} = \{HH, HT\}$

$H_2 = \{2nd\ toss\ is\ a\ head\} = \{HH, TH\}$

$D = \{the\ two\ tosses\ produced\ different\ results\} = \{HT, TH\}$

$\Omega = \{HH, HT, TH, TT\}$

(a) $P(H_1) = P(H_2) = 1/2$

$P(H_1 \cap H_2) = P(\{HH\}) = 1/4 = P(H_1)P(H_2)$

$\rightarrow H_1$ and $H_2$ are unconditionally independent

(b) $P(H_1|D) = \frac{P(H_1 \cap D)}{P(D)} = \frac{1/4}{1/2} = 1/2$

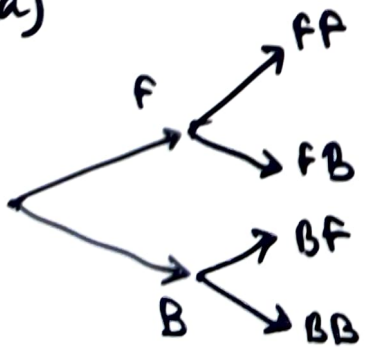
$P(H_2|D) = \frac{P(H_2 \cap D)}{P(D)} = \frac{1/4}{1/2} = 1/2$

$P(H_1 \cap H_2|D) = \frac{P(H_1 \cap H_2 \cap D)}{P(D)} = 0$

$\because H_1 \cap H_2 \cap D = \phi$

$\rightarrow H_1, H_2$ are not conditionally independent given D.

2. (a)




F: event of taking a step forward

B: event of taking a step backward

$P(F) = p$, $P(B) = 1-p$

Assuming subsequent steps are independent

P(same place after two steps) = P({FB, BF})

= 2p(1-p)

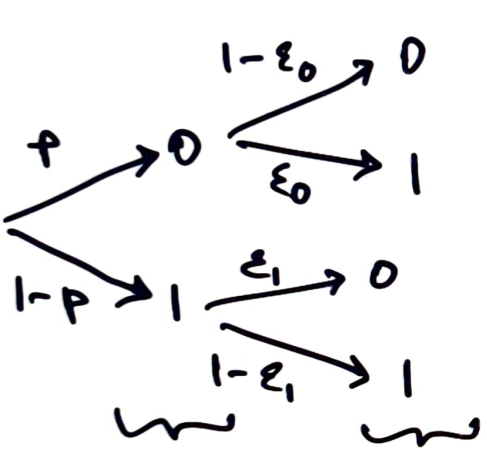
(b) P(one step forward after 3 steps) = P({FFB, FBF, BFF})

= 3p²(1-p)

(c) P(step 1 = F | one step forward after 3 steps)

$$= \frac{P(\{FFB, FBF\})}{P(\{FFB, FBF, BFF\})} = 2/3$$

3.




(a) P(correct reception)

= P({00, 11})

= P(0)P(0|0) + P(1)P(1|1)

= $p(1-\varepsilon_0) + (1-p)(1-\varepsilon_1)$

(b) P(101 correctly received)

= $(1-\varepsilon_1)^3\varepsilon_0$

(c) P(0 is correctly decoded | 0 is transmitted)

= P({000, 100, 010, 001})

= $(1-\varepsilon_0)^3 + 3(1-\varepsilon_0)^2\varepsilon_1 = (1-\varepsilon_0)^2(1-\varepsilon_0+3\varepsilon_1)$

(d) 0: 0 is transmitted → {000, 100, 010, 001, 110, 101, 011, 111}

1: 1 is transmitted → same as above

P(101) = P(101|0)·P(0) + P(101)·P(1)

$= (1-\varepsilon_0)\varepsilon_0^2 p + (1-\varepsilon_1)^2 \varepsilon_1 (1-p)$

$\Rightarrow P(0|101) = \dfrac{P(0 \cap 101)}{P(101)} = \dfrac{P(101|0)P(0)}{P(101)}$

$= \dfrac{(1-\varepsilon_0)\varepsilon_0^2 p}{(1-\varepsilon_0)\varepsilon_0^2 p + (1-\varepsilon_1)^2 \varepsilon_1 (1-p)}$

4. (a) $P(A|A) = \dfrac{P(A)}{P(A)} = 1 \neq P(A).P(A)$ if $A \subset \Omega$

A is independent of itself if $A = \Omega$ or $\phi$

(b) A and B are independent $\Rightarrow P(A \cap B) = P(A)P(B)$

$A = (A \cap B) \cup (A \cap B^c) \Rightarrow P(A) = P(A \cap B) + P(A \cap B^c)$

$\Rightarrow P(A \cap B^c) = P(A) - P(A \cap B)$

$= P(A) - P(A)P(B) = P(A)(1-P(B))$

$= P(A)P(B^c)$

$\Rightarrow$ A and $B^c$ are independent

(c) A, B, C are independent events

$P(A \cap B) = P(A)P(B)$, $P(B \cap C) = P(B)P(C)$, $P(C \cap A) = P(C)P(A)$

$P(A \cap B \cap C) = P(A)P(B)P(C)$

Now $P(A \cap B|C) = \dfrac{P(A \cap B \cap C)}{P(C)} = P(A)P(B) = P(A|C)P(B|C)$

$\Rightarrow$ A and B are conditionally independent given C.